coursera

Depressive and Bipolar Disorders

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# Brain Stimulation Therapies

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In cases of treatment resistant depression and bipolar disorder, clinicians can use **brain-stimulation therapies** to alter brain activity and alleviate symptoms of depression or mania. Because these disorders present differently in different patients, therapists formulate a plan for using these techniques based on the symptoms of each individual patient. Some techniques, such as electroconvulsive therapy, have been used for decades, while others, such as transcranial magnetic stimulation (TMS), deep brain stimulation (DBS), and vagus nerve stimulation (VNS), are much newer (Leuchter et al., 2013, Drobisz & Damborská, 2019). Other techniques that were used more widely in the past, such as psychosurgery, are now used in very few, severe cases of treatment-resistant depression.

## Types of Brain Stimulation

### Electroconvulsive Therapy

**Electroconvulsive therapy (ECT)** is a treatment in which a seizure is induced in the patient by passing a controlled, low-dose electric current through one or both temples. It is the oldest and the most well-known therapeutic brain stimulation technique, and it has been used to treat psychological disorders for more than 80 years (Gazdag & Ungvari, 2019). The use of ECT declined during the 1970s and 1980s, after the introduction of antidepressant medications, and it is now mainly used for severe, medication-resistant cases of major depression or bipolar disorder (Espinoza & Kellner, 2022).

Modern ECT is administered as an in- or out-patient setting 2 or 3 times a week for 2-4 weeks (Espinoza & Kellner, 2022). To ensure the safety of patients, they are closely monitored before, during, and after the procedure. First, patients are anesthetized to address any discomfort and given muscle relaxants to reduce convulsions. Electrodes are placed on the patient's forehead with gels to prevent electrical burns. Trained medical staff monitor the patients' brain, cardiac, and respiratory activity during the procedure (Kameg & Kameg, 2021; Swartz & Nelson, 2005). Patients often have a period of tapering ECT session and continue to use antidepressant medication after their initial ECT sessions to prevent relapse (Espinoza & Kellner, 2022). While the neurological mechanisms of how ECT works remain unclear, there is evidence to suggest that ECT may affect brain structure (e.g., increases in gray-matter in the hippocampus and amygdala) and neurotransmitter systems (Espinoza & Kellner, 2022).

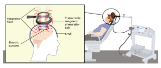
ECT treatment is associated with concerning side effects, such as cognitive problems and deficits in autobiographical (e.g., long-term) memory. Yet, many of the cognitive problems decrease over time (Dominiak et al., 2021). For example, Marilyne Landry and colleagues (2021) conducted a review and meta-analysis of the most common side effects following ECT from 91 published studies. They found small to medium size decreases in autobiographical memory, verbal fluency, and verbal memory during short-term measurements (1-28 days following treatment), yet there was improvement in nearly all areas of cognitive functioning during long-term measurements (over 1 month following treatment). In addition, the cognitive side effects vary based on the positioning of the electrodes on the patient's head, along with the type and dose of electrical stimulation used (Dominiak et al., 2021).

Concerns about the side effects along with the negative portrayal of ECT in movies has stigmatized the use of this treatment (Kellner et al., 2020). Yet, despite the drawbacks of ECT, such as hospitalization, use of a general anesthetic, induction of seizure, and possible cognitive impairments, ECT has a high rate of effectiveness (Dominiak et al., 2021; Minichino et al., 2012). For example, Mohn and Rishovd Rund (2019) found that remission rates increased over time from 26.9% at 6 weeks following treatment to 62.9% of patients at 2 years post treatment.

### Repetitive Transcranial Magnetic Stimulation (rTMS)

**Repetitive transcranial magnetic stimulation or rTMS** is a non-invasive procedure used for treatment-resistant depression. With rTMS, a handheld electromagnet is placed close to the surface of the head at the site of interest as shown in the *rTMS* image. The repetitive activity when the magnet is switched on and off stimulates specific brain regions under the magnet. The disruption is similar to a localized seizure. Studies suggest that rTMS helps regulate the connectivity in parts of the frontal lobe, which is involved in emotion regulation (Kang et al., 2016).

***rTms***



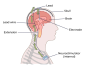
*Note.* The figure illustrates a person receiving rTMS. The person is lying back in a chair and the transcranial magnetic stimulation (TMS) coil, which is connected to a stimulus generator, is placed close to the surface of the head. The TMS coil produces a magnetic field which passes through the skull and produces an electric current in a localized area of the brain.

Treatments typically occur 5 days a week and last 4-6 weeks. When the magnet is turned on, the patient may hear some loud clicking noises and feel tingling or see a bit of movement in their hand with each click. The only major side effect is people often report having a headache that goes away in a few hours. rTMS doesn't work for everyone, but between 20 to 30% of people see significant improvement in or remission of their symptoms (Somani & Kar, 2019).

### Deep Brain Stimulation (DBS)

**Deep brain stimulation or DBS** is a neurosurgical procedure where electrodes are implanted in the brain to deliver small electrical impulses to treat treatment-resistant depression. First, electrodes are placed in subcortical regions of the brain associated with depressive symptoms, such as the nucleus accumbens which is associated with positive emotions. Next, the pulse generator is surgically implanted just below the collarbone with wires implanted under the skin and connected to the electrode lead. The general placement is illustrated in the *DBS* image. The pulse generator contains a battery pack that delivers a gentle electrical current to stimulate neurons near the electrodes. An external handheld device allows physicians to adjust the pulse generator's settings.

***DBS***



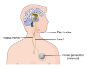
*Note.* An illustration of the chest and head of an individual receiving deep brain stimulation (DBS). A neurostimulator is placed internally just below the collarbone with wires implanted under the skin and connected to an extension and electrode lead wire, which move up along the outside of the brain. The final connection is with an electrode located in subcortical regions of the brain. The electrical current stimulates neurons near the electrode.

Deep brain stimulation is considered an extreme treatment option because of the inherent risks that accompany surgical procedures. DBS is effective in between 56% and 80% of patients with treatment-resistant depression (Riva-Posse et al., 2018, Y. Wu et al., 2021). However, in order to maintain the improvement patients must continue the stimulation. Research showed that patients who turned off the stimulation experienced a return of their depression symptoms, and turning the devices back on alleviated the returned symptoms (Delaloye & Holtzheimer, 2014; Holtzheimer et al., 2017).

### Vagus Nerve Stimulation (VNS)

**Vagus nerve stimulation or VNS** tries to manipulate how the brain functions through an indirect method—stimulation of the vagus nerve. The vagus nerve is the longest nerve in the autonomic nervous system, running from the brain through the neck to the chest and the stomach; it includes both motor and sensory nerves. Stimulation is achieved through an electrode implanted near the vagus nerve, which is activated by a pulse generator device implanted under the skin on the patient's chest. The pulse generator is used to intermittently stimulate the vagus nerve on the left side of the neck. The *VNS* image illustrates the placement of these devices.

***VNS***



*Note.* An illustration of the chest and head of an individual receiving vagus nerve stimulation (VNS), showing the brain and the vagus nerve moving down from the brain through the neck to the chest. A pulse generator device, under the skin on the patient's chest, is connected to a lead wire and an electrode located near the vagus nerve. The pulse generator is used to intermittently stimulate the electrode and the nearby vagus nerve.

In contrast to rTMS and DBS, which directly stimulate brain areas, VNS is considered a "bottom-up" procedure in which stimulation of the peripheral nerves activates the central nervous system. The exact mechanism through which VNS alleviates depression is currently unknown (Yap et al., 2020).

The effectiveness of VNS in reducing the symptoms of depression increases based on the length of VNS treatment. For example, the response rate, in which individuals reach at least a 50% reduction in symptoms, at 3 months was 37%, but increased to 53% at 12 months (Schlaepfer et al., 2008).

VNS is used infrequently due its limited availability, invasiveness, and high cost. Non-invasive versions, in which a stimulator is placed in the ear like a hearing aid, are increasing in use, however, the effectiveness of these versions has not yet been established (Austelle at al., 2022; Howland, 2014).

## Psychosurgery

Psychosurgery is the most drastic measure that is recommended only to patients with the most severe depression who can't be helped by any other method. Psychosurgery is the treatment of a mental disorder by surgical removal or destruction of selective brain areas. The most well-known example of psychosurgery is prefrontal lobotomy, historically used particularly for schizophrenia but also a variety of other disorders. Before dismissing this out of hand, today's psychosurgery is not like the lobotomies performed 80 to 100 years ago that destroyed large areas of the brain. Now, surgeons can focus on specific areas, such as the anterior cingulate, destroy very little tissue, and see positive outcomes in a third to a half of patients (Volpini et al., 2017). Recent studies reveal that deep brain stimulation has similar efficacy rates and surgical risks as psychosurgery (Volpini et al., 2017). However, unlike deep brain stimulation, psychosurgery is irreversible and any undesired results are permanent. Thus, psychosurgery is only considered as a last resort when all other techniques have failed to bring relief and the patient's quality of life is intolerable in the absence of an effective intervention.

## Comparison: Brain Stimulation Therapies

The following table provides a summary and comparison of the brain stimulation techniques. Notice that while all of the methods are utilized for treatment-resistant depression, they vary in several ways. For example, only DBS and VNS require invasive surgery, while only DBS provides direct stimulation of the brain. Both rTMS and DBS provide targeted stimulation at specific sites in the brain.

| Brain stimulation therapies | Invasive surgery required | Direct stimulation of the brain | Targeted stimulation at specific sites | Used for treatment-resistant depression |
|---|---|---|---|---|
| Electroconvulsive therapy (ECT) | N/A | N/A | N/A | X |
| rTMS | N/A | N/A | X | X |
| Deep Brain Stimulation (DBS) | X | X | X | X |
| Vagus Nerve Stimulation (VNS) | X | N/A | N/A | X |

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